

ulms-highgrade-pulm-mets-tp53-smarc b1-hla-a2-w9t4

Selected general biomarker report — biomarker screening coverage and gaps

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Selected general biomarker report

Which biomarkers on the selected panel this patient has been tested for, and which have never been measured. The panel covers targets established in this tumor type, targets relevant across a subset of tumor types, and biomarkers that predict benefit regardless of where the tumor started. An unmeasured biomarker below is a gap in the record, not a prediction that the patient carries it.

In scope: 18 biomarkers. **No usable result on file:** 15. **Measured but not decision-grade:** 2. **Already established:** 1. **Assessed and set aside:** 63.

Eighteen biomarkers were in scope for this tumor. Fifteen have no result at all, two were measured short of decision grade, one is settled. Four gaps are essential: MSI/dMMR, tumor mutational burden, NTRK fusion, and homologous recombination with BRCA status. The other 63 panel entries were assessed against this tumor and set aside, with reasons at the foot of the page.

Plasma against tissue is the distinction doing the work. The Guardant tumor fraction was never supplied, so its non-detections exclude nothing. NTRK and HR/BRCA are logged as measured but not hardened: a plasma negative exists for each. MSI and TMB stayed unmeasured: the plasma assay returned no call, so there is nothing to harden.

HLA class I typing is the one settled row, and typing is not eligibility. A confirmed A*02:01 turns MAGE-A4, PRAME and NY-ESO-1 into live expression questions, all three off one archival block. Leiomyosarcoma runs well below synovial sarcoma for MAGE-A4 and NY-ESO-1; PRAME is the likeliest of the three.

Three bundles and one blood draw carry nearly every actionable row: comprehensive tissue DNA plus RNA profiling, an IHC block led by INI1, the cancer/testis stain set, and a germline HR panel. Nothing here needs a fresh biopsy, and no biopsy or cost preference was recorded at intake. Every tissue order waits on release of the September 2024 outside-institution block, the rate-limiting step; with LMS-04 pending, ordering in series costs weeks. INI1 is the stain that would move the plasma SMARCB1 finding from hypothesis to actionable, against a hard prior: 170 consecutive uterine smooth-muscle tumors all retained SMARCB1. Eight rows hand off to the target-validation track.

Most of this testing will come back negative. What justifies running it is the size of the option a positive would open, not any expectation that one is coming.

Gaps: biomarkers with no usable result on file

Microsatellite instability / mismatch-repair deficiency (MSI-H / dMMR)

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

MSI-H and dMMR are read the same way whatever the primary site. Nothing in this record speaks to mismatch repair, and she becomes a checkpoint-blockade candidate the moment the planned first-line regimen stops working.

Mismatch-repair status is still unassessed: the plasma assay returned no MSI call, and no stain or PCR has been run on either specimen. Uterine leiomyosarcoma is nearly always mismatch-repair proficient, so a negative is the

expected result, and the case for testing is that four stains on a block already being cut for INI1 close a tumor-agnostic PD-1 indication that would otherwise sit open at progression. Loss of any one protein would also carry Lynch implications for a woman in her thirties and for her first-degree relatives.

Recommended assay. MMR protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue (IHC · Archival FFPE; the September 2024 hysterectomy specimen has more tissue to spare, the 2026 lung core reflects current disease · 3-5 days)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3

A positive result would open.

- pembrolizumab (PD-1 inhibitor · tumor-agnostic approval · NCT02628067)
- dostarlimab (PD-1 inhibitor · tumor-agnostic approval · NCT02715284)

Next step. Order now

References. nct:NCT02628067; nct:NCT02715284

Tumor mutational burden, high (TMB-H)

Priority: essential **Scope:** Tumor-agnostic **Status:** Not measured

TMB at or above 10 mutations per megabase supports pembrolizumab whatever the primary site. The Guardant plasma assay did not report a TMB value, so TMB here is unknown rather than low.

TMB is unreported: the plasma assay did not return one, and a plasma TMB at an unstated tumor fraction would not have been usable anyway. Leiomyosarcoma is copy-number driven and typically low-TMB, so a high value would be a surprise, but the tissue panel that reports it is the same order that settles MSI, the fusions, the HR genes, and whether the plasma TP53 call is tumor-derived. Ask explicitly for a validated comprehensive assay with a mutations-per-megabase figure, because a number off a small hotspot panel does not satisfy the indication.

Recommended assay. Validated comprehensive genomic profiling panel on tissue, reporting TMB in mutations per megabase (NGS_panel · Archival FFPE; the September 2024 uterine resection is preferred for tumor content and volume · 2-3 weeks)

Bundles with. Comprehensive tissue DNA plus RNA genomic profiling on archival tumor tissue, the single order that also returns TMB, MSI, NTRK and RET fusions, BRAF codon 600, the HR genes, the RB1 / ATRX / MED12 backbone of this histology, and tissue confirmation of the plasma TP53 and SMARCB1 calls

A positive result would open.

- pembrolizumab (PD-1 inhibitor · tumor-agnostic approval · NCT02628067)

Next step. Order now

References. nct:NCT02628067

B7-H3 (CD276) protein expression

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

B7-H3 sits on tumor cells and tumor vasculature across soft tissue sarcoma: 97% of specimens positive and about 69% high in a 153-patient series covering 15 subtypes, leiomyosarcoma among them. Of the surface targets on the panel, this is the one most likely to be positive here.

B7-H3 has still not been stained, and it carries the best odds of a positive result of any surface target in scope. Nothing is approved against it, so a positive buys trial access rather than a prescription, and the ADC, CAR-T and bispecific programs generally require documented expression at screening. Adding it to the block already being cut for INI1 and the MMR stains keeps a later trial referral from stalling while someone hunts for tissue.

Recommended assay. B7-H3 (CD276) IHC on tumor tissue (IHC · Archival FFPE; the 2026 lung core reflects current disease, the 2024 resection has more tissue and reads heterogeneity better · 1 week)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3

A positive result would open.

- ifinatamab deruxtecan (B7-H3-directed ADC · trial only)
- enoblituzumab (mAb · trial only)
- vobramitamab duocarmazine (ADC · trial only)

Next step. Bundle with planned testing

References. pmid:39478506

MAGE-A4 expression (HLA-A*02:01-restricted)

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

MAGE-A4 carries the first approved TCR-T therapy for a solid tumor, and this patient clears its restriction gate: allele-level typing confirms HLA-A*02:01. Expression has never been tested, and it is the only remaining half of the screening question.

The expensive half of this question is already answered: she is HLA-A*02:01 positive, so MAGE-A4 expression is the only thing standing between her and a class of agents she is a plausible candidate for. Expression in leiomyosarcoma runs well below the 70 to 80 percent seen in synovial sarcoma, so a negative is a realistic and useful outcome, and even a positive would not make her eligible for afamitresgene autoleucel, whose licence covers synovial sarcoma alone. The realistic route is a MAGE-A4 solid-tumor TCR-T trial, and the stain is what decides whether that door is open or shut.

Recommended assay. MAGE-A4 expression by IHC or RT-PCR on tumor tissue, scored per the candidate program's screening assay (IHC · Archival FFPE; RT-PCR tolerates lower tumor content than IHC scoring · 1-2 weeks)

Bundles with. Cancer/testis antigen IHC set (MAGE-A4, PRAME, NY-ESO-1) on one archival block, now unconditional because HLA-A*02:01 is confirmed

A positive result would open.

- afamitresgene autoleucel (MAGE-A4-directed TCR-T · approved, other tumor type)
- uzatresgene autoleucel (MAGE-A4-directed TCR-T · trial only)

Next step. Order if tissue available

References. pmid:38554725; pmid:36624315

PRAME expression (HLA-A*02:01-restricted)

Priority: high **Scope:** Relevant across a tumor subset **Status:** Not measured

PRAME is expressed in a subset of sarcomas including leiomyosarcoma, and the ImmTAC and TCR-T programs behind it enrol HLA-A*02:01-positive patients, which she is. Antigen expression has never been measured.

PRAME is now a straight expression question, because the HLA gate that used to make it unassessable is answered. Of the three cancer/testis antigens in scope it has the most credible chance of being positive in this histology, and both programs behind it enrol HLA-A*02:01-positive, PRAME-expressing solid tumors, so a positive buys trial access rather than a prescription. Reported expression in leiomyosarcoma runs below synovial sarcoma, so a negative stain would not be surprising and would close the axis cleanly.

Recommended assay. PRAME expression by validated IHC or RT-PCR at the threshold the candidate protocol specifies (IHC · Archival FFPE; RT-PCR tolerates lower tumor content than IHC scoring · 1-2 weeks)

Bundles with. Cancer/testis antigen IHC set (MAGE-A4, PRAME, NY-ESO-1) on one archival block, now unconditional because HLA-A*02:01 is confirmed

A positive result would open.

- brenetafusp (PRAME-directed bispecific TCR (ImmTAC) · trial only)
- IMA203 (PRAME-directed TCR-T · trial only)

Next step. Order if tissue available

References. pmid:40205198

BRAF V600E mutation

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

BRAF V600E is actionable across nearly all primary sites, colorectal excepted. It is a rarity in leiomyosarcoma, and it belongs on the list because any comprehensive tissue panel reports codon 600 whether or not anyone thought to ask.

No BRAF result is on file and the prior probability in uterine leiomyosarcoma is very low. No separate order is warranted: codon 600 is covered by the comprehensive tissue panel already recommended for TMB, MSI and the HR genes. Non-V600E class II and III BRAF alterations, should any turn up, do not carry this indication.

Recommended assay. Tumor NGS panel covering BRAF codon 600 (NGS_panel · Archival FFPE · 2-3 weeks)

Bundles with. Comprehensive tissue DNA plus RNA genomic profiling on archival tumor tissue, the single order

that also returns TMB, MSI, NTRK and RET fusions, BRAF codon 600, the HR genes, the RB1 / ATRX / MED12 backbone of this histology, and tissue confirmation of the plasma TP53 and SMARCB1 calls

A positive result would open.

- dabrafenib plus trametinib (BRAF inhibitor plus MEK inhibitor · tumor-agnostic approval · NCT02034110)

Next step. Bundle with planned testing

References. nct:NCT02034110

HER2 protein overexpression (IHC 3+)

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

HER2 IHC 3+ supports trastuzumab deruxtecan irrespective of primary site. Strong HER2 staining is uncommon in leiomyosarcoma, but nothing on file addresses it and the stain is routine.

HER2 protein expression is untested. Only 3+ carries the tumor-agnostic indication and 3+ is unlikely in uterine leiomyosarcoma, so a 1+ or 2+ result should be reported as HER2-low and read as a weaker, less reliable signal rather than as a positive. A block is being cut for other stains anyway, which makes adding HER2 cheap and settles the question either way. The ERBB3 G914E variant on the plasma report sits at 0.9% VAF, near the assay limit of detection, and is not a reason to pursue the HER family.

Recommended assay. HER2 IHC on tumor tissue (IHC · Archival FFPE block · 1 week)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3; ERBB2 copy number also comes off the comprehensive tissue NGS panel

A positive result would open.

- trastuzumab deruxtecan (HER2-directed ADC · tumor-agnostic approval · NCT04482309)

Next step. Bundle with planned testing

References. nct:NCT04482309

RET gene fusion

Priority: medium **Scope:** Tumor-agnostic **Status:** Not measured

The RET fusion indication ignores primary site. RET fusions in uterine leiomyosarcoma are rare to the point of being anecdotal, but the RNA fusion assay ordered for NTRK answers RET in the same run.

RET fusion is untested and will almost certainly be negative in this histology, and the plasma report on file does not enumerate a RET result either way. It earns a row because it costs nothing extra once an RNA fusion panel is ordered for NTRK, and because a positive would put an oral targeted agent ahead of further cytotoxic chemotherapy. A RET point mutation is a different finding and does not carry this indication.

Recommended assay. RNA-based fusion NGS panel covering RET (NGS_panel · Archival FFPE · 2-3 weeks)

Bundles with. Comprehensive tissue DNA plus RNA genomic profiling on archival tumor tissue, the single order that also returns TMB, MSI, NTRK and RET fusions, BRAF codon 600, the HR genes, the RB1 / ATRX / MED12 backbone of this histology, and tissue confirmation of the plasma TP53 and SMARCB1 calls

A positive result would open.

- selpercatinib (selective RET inhibitor · tumor-agnostic approval · NCT03157128)

Next step. Bundle with planned testing

References. nct:NCT03157128

FAP (fibroblast activation protein) expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

FAP is the panel's stromal target, and mesenchymal tumors carry it on the malignant cells as well as on associated fibroblasts, which is why sarcoma shows up so reliably on FAPI PET. The panel's expression evidence comes from carcinoma stroma, so read sarcoma relevance as supported by imaging rather than by an IHC series in this histology.

FAP is untested, and no FAP-directed agent is approved anywhere, so a positive would matter only if a radioligand or bispecific trial were within reach. FAP IHC is not a routine clinical stain and most laboratories do not offer it, which makes speculative testing a poor use of a block that several higher-yield stains are already queued on. Revisit if a FAP-directed trial becomes a live option, at which point FAPI PET is usually the practical screening route and spends no tissue.

Recommended assay. FAP IHC on archival tissue, or FAPI PET where a FAP-directed program requires it (IHC · Archival FFPE · 1-2 weeks)

Bundles with. The archival IHC block if a FAP-directed program comes into play; otherwise the trial's own screening assay

A positive result would open.

- 177Lu-FAP-2286 (radioligand therapy · trial only)
- RO7300490 (bispecific (FAP-4-1BBL) · trial only)
- simlukafusp alfa (FAP-IL2v immunocytokine · trial only)

Next step. Defer

References. pmid:32601975

HER2 (ERBB2) protein expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

HER2 is the most heavily binder-backed surface target on this panel, and an IHC 3+ result carries a tumor-agnostic indication whatever the primary site. Strong staining is uncommon in uterine leiomyosarcoma, and nothing on file addresses it either way.

This is the selected-panel entry for HER2 and it covers the same ground as the tumor-agnostic HER2 row in this survey rather than adding a second order. HER2 protein has not been stained; only IHC 3+ carries the tumor-agnostic indication, so a 1+ or 2+ result should be reported as HER2-low and read as a weaker signal rather than as a positive. The stain rides on the archival block already being cut for the other IHC work.

Recommended assay. HER2 IHC on tumor tissue (IHC · Archival FFPE block · 1 week)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3

A positive result would open.

- fam-trastuzumab deruxtecan-nxki (HER2-directed ADC · tumor-agnostic approval)
- trastuzumab (mAb · approved, other tumor type)
- zanidatamab (bispecific · approved, other tumor type)

Next step. Bundle with planned testing

NY-ESO-1 (CTAG1B) expression, HLA-A*02:01-restricted

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

NY-ESO-1 is the third A*02:01-restricted cancer/testis antigen with a clinical-stage TCR-T program, and the restriction gate is satisfied. Its expression evidence sits in synovial sarcoma and myxoid liposarcoma rather than leiomyosarcoma, so the prior here is low.

NY-ESO-1 is an expression question now rather than an unassessable one, since HLA-A*02:01 is confirmed, but the prior probability in leiomyosarcoma is low. Letetresgene autoleucel is being developed in synovial sarcoma and myxoid/round cell liposarcoma, so even a positive stain would point at a solid-tumor cohort rather than a trial written for this histology. Add it to the cancer/testis block being cut for MAGE-A4 and PRAME rather than ordering it on its own.

Recommended assay. NY-ESO-1 expression by IHC or RT-PCR, run alongside MAGE-A4 and PRAME on the same block (IHC · Archival FFPE; RT-PCR is the more forgiving option at low tumor content · 1-2 weeks)

Bundles with. Cancer/testis antigen IHC set (MAGE-A4, PRAME, NY-ESO-1) on one archival block, now unconditional because HLA-A*02:01 is confirmed

A positive result would open.

- letetresgene autoleucel (NY-ESO-1-directed TCR-T · trial only · NCT03967223)

Next step. Bundle with planned testing

References. nct:NCT03967223

PD-L1 protein expression

Priority: medium **Scope:** Relevant across a tumor subset **Status:** Not measured

PD-L1 gates no approval in soft tissue sarcoma, so this is supporting rather than decisive. It stays in scope because the pulmonary lesions were read as infiltrative and possibly inflammatory, which makes the immune phenotype a question someone will raise.

PD-L1 has not been stained and cannot be read off a plasma assay. In sarcoma it neither opens nor closes a labelled indication, so treat the result as supporting information for trial screening and for weighing any checkpoint-blockade proposal rather than as a decision point. If it is stained, use the lung metastasis and match the clone and cutoff to whichever agent is under consideration, because a result generated for one companion diagnostic does not transfer to another indication.

Recommended assay. PD-L1 IHC using the clone and scoring system matched to the intended agent (IHC · Archival FFPE; the 2026 lung core is the more informative specimen for current immune phenotype · 1 week)

Bundles with. Single archival block cut for the IHC set: INI1/SMARCB1, MMR (MLH1, PMS2, MSH2, MSH6), PD-L1, HER2 and B7-H3

A positive result would open.

- PD-1 / PD-L1 inhibitors (checkpoint inhibitor · trial only)

Next step. Bundle with planned testing

AXL receptor tyrosine kinase expression

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not measured

AXL marks mesenchymal and EMT-high tumor cells, which is the constitutive state of a sarcoma rather than an acquired one, and it tracks with chemotherapy resistance. The panel's expression evidence comes from carcinomas, so read this as plausible in leiomyosarcoma rather than established.

AXL has not been assessed, and no AXL-directed agent is approved: enapotamab vedotin and tilvestamab both stopped at completed early-phase studies. Testing would not change anything she can access now or at progression. Revisit only if an AXL-directed program opens with an expression-based entry criterion.

Recommended assay. AXL IHC on tumor tissue (IHC · Archival FFPE · 1-2 weeks)

Bundles with. Would ride on the archival IHC block if an AXL-directed program required documented expression

A positive result would open.

- enapotamab vedotin (ADC · trial only)
- tilvestamab (mAb · trial only)

Next step. Defer

MAGE-C2 (MAGEC2) expression, HLA class I presented

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not measured

MAGE-C2 is the one cancer/testis antigen on this panel whose expression list names sarcoma alongside

melanoma and hepatocellular carcinoma, and the confirmed HLA class I typing keeps HLA-presented antigens assessable rather than moot. Expression has never been measured.

MAGE-C2 sits behind the three cancer/testis antigens already queued for staining and has one academic TCR-T programme behind it, developed in melanoma and head and neck cancer rather than sarcoma, so a result would not open anything she could reach now. It earns a row because the panel reports expression in sarcoma and her HLA typing keeps the antigen class live. If the MAGE-A4, PRAME and NY-ESO-1 stains are all negative and a MAGE-C2 protocol opens, this is the next antigen to ask about.

Recommended assay. MAGE-C2 expression by IHC or RT-PCR, only if a MAGE-C2-directed protocol becomes reachable (IHC · Archival FFPE · 1-2 weeks)

Bundles with. Would ride on the cancer/testis antigen block already being cut for MAGE-A4, PRAME and NY-ESO-1

A positive result would open.

- MC2 TCR-T (MAGE-C2-directed TCR-T · trial only)

Next step. Defer

PSMA (FOLH1) expression, tumor cell and tumor neovasculature

Priority: low **Scope:** Relevant across a tumor subset **Status:** Not measured

PSMA is expressed on the neovasculature of many solid tumors as well as on prostate epithelium, and that vascular uptake is what a completed phase 1 radioligand study used to select soft-tissue sarcoma patients. This case's own trial screen surfaced that study, so the target is decision-adjacent here despite the panel's prostate framing.

PSMA has not been imaged, and the one sarcoma study that would have used the scan to select patients is completed and was ranked unavailable for her in this case's own screen. Imaging now would produce a result with nowhere to go, and PSMA PET is an expensive scan to run on that basis. Revisit only if a PSMA-selected sarcoma protocol opens within reach.

Recommended assay. PSMA PET/CT, only if a PSMA-selected sarcoma protocol becomes available to her (imaging · None; imaging spends no tissue · 1-2 weeks to schedule and report)

A positive result would open.

- Lutetium Lu 177 vipivotide tetraxetan (radioligand · approved, other tumor type)
- Lu-177-ITG-PSMA-1 with Ga-68-PSMA-11 PET selection (radioligand · trial only · NCT05420727)

Next step. Defer

References. nct:NCT05420727

Measured, but not to decision resolution

A result exists for each of these, but not at the resolution a treatment decision needs. The workup that would close each gap is carried into the Target validation paths report.

NTRK1 / NTRK2 / NTRK3 gene fusion

Priority: essential

On file. profile.json:biomarkers[13] NTRK fusion: 'not settled by the Guardant plasma panel (fusion detection in plasma is insensitive at low tumor fraction)', confirmation_status 'ngs_pending'; no RNA-based fusion panel and no tissue DNA NGS with fusion calling on file; profile.json:organ_function.note_ctdna_tumor_fraction: plasma tumor fraction was not supplied, so non-detection on the plasma assay is non-exclusionary

Gap. All that is on file is non-detection on a plasma DNA assay run at an unstated tumor fraction. Plasma fusion calling is insensitive when shedding is low, and DNA capture misses breakpoints sitting in large NTRK introns. An RNA-based fusion panel on archival tissue, covering NTRK1, NTRK2 and NTRK3, is what would let a negative be read as a negative.

A plasma assay reported no NTRK fusion, which is not the same thing as a negative: tumor fraction was never supplied, and plasma DNA fusion calling misses breakpoints an RNA assay would catch. Prevalence in uterine leiomyosarcoma is very low, so a true negative is the expected outcome, and what justifies the assay is that a positive turns a chemotherapy-only landscape into an oral TRK inhibitor with high and durable response rates. It rides on the same tissue order that answers RET, BRAF, TMB and MSI, so it adds little beyond the tissue itself.

Homologous recombination deficiency / BRCA1 / BRCA2 pathogenic variant

Priority: essential

On file. profile.json:biomarkers[12] Homologous recombination status incl. BRCA2: 'not settled by the Guardant plasma panel (no BRCA1/2 alteration reported, but plasma sensitivity is limited at low tumor fraction and HRD signatures require tissue)', confirmation_status 'ngs_pending'; no tissue HRD score and no germline multigene panel on file; profile.json:organ_function.note_ctdna_tumor_fraction: plasma tumor fraction was not supplied, so non-detection on the plasma assay is non-exclusionary

Gap. The plasma panel reported no BRCA1/2 alteration, at an unstated tumor fraction and with no genomic-instability score, and no germline testing has been done. Closing this needs tissue NGS across the HR genes with an HRD score, plus a germline multigene panel on blood.

The plasma assay does not settle homologous-recombination status: an absent BRCA1/2 call at unknown tumor fraction is weak evidence, and an HRD signature cannot be derived from that assay at all. Roughly a tenth of uterine leiomyosarcomas carry an HR pathway mutation with a larger fraction functionally deficient by RAD51 assay, and a pathogenic finding would put olaparib plus temozolomide and PARP-inhibitor trials on the table for later lines. Order the germline panel alongside the tumor test rather than after it, since somatic-only testing leaves both her own risk and her relatives' screening unanswered.

Already established

- **HLA class I genotype** — profile.json:biomarkers[4] HLA class I typing (allele-level): A02:01, A03:01; B39:01, B

44:02; C05:01, C12:03, confirmation_status 'confirmed'; profile.json:targetable_features[2] carries the A*02:01 result as an opened axis pending antigen expression

Remaining panel biomarkers

The other 63 of 72 reference-panel targets, each assessed against this case. None of them changes what to order today.

- **Alkaline phosphatase, placental-like (germ-cell) (ALPPL2) protein expression** — Onco-testicular antigen re-expressed in pancreatic, mesothelial, ovarian and germ-cell tumors; no reported expression in uterine smooth-muscle tumors, and the panel lists no binder program behind it.
- **Anthrax toxin receptor 1 (TEM8) (ANTXR1) protein expression** — Reported on tumor vasculature and on triple-negative breast, neuroendocrine and small-cell lung carcinoma cells; no reported expression in uterine leiomyosarcoma, and the programs behind it are preclinical.
- **B-cell maturation antigen (BCMA) (TNFRSF17) protein expression** — BCMA is restricted to plasma cells and is the canonical multiple myeloma target; no lineage basis in a soft-tissue sarcoma.
- **B-lymphocyte antigen CD19 (CD19) protein expression** — Pan-B-cell antigen behind the CAR-T and bispecific programs for lymphoid leukemia and lymphoma; a soft-tissue sarcoma carries no B-cell lineage to target.
- **C-type lectin domain family 12 member A (CLL-1) (CLEC12A) protein expression** — Antigen of acute myeloid leukemia blasts and leukemic stem cells; no lineage basis in a soft-tissue sarcoma.
- **CD70 antigen (CD70) protein expression** — Constitutively expressed in clear-cell renal cell carcinoma and in several lymphomas; no reported expression in uterine smooth-muscle tumors.
- **CUB domain-containing protein 1 (CDCP1) protein expression** — Up-regulated in breast, ovarian, oral squamous, pancreatic and lung carcinomas; an epithelial target with no reported expression in uterine leiomyosarcoma.
- **Cadherin-17 (LI-cadherin) (CDH17) protein expression** — Lineage-restricted to gastrointestinal epithelium and expressed in colorectal, gastric and pancreatic adenocarcinoma; no reported expression in uterine smooth-muscle tumors.
- **Cadherin-3 (P-cadherin) (CDH3) protein expression** — P-cadherin marks basal-like breast, pancreatic, gastric, ovarian and endometrial carcinoma; endometrial carcinoma is an epithelial tumor of the uterus, not this smooth-muscle sarcoma.
- **Cadherin-6 (K-cadherin) (CDH6) protein expression** — Over-expressed in papillary renal, ovarian, biliary and thyroid carcinoma; no reported expression in uterine leiomyosarcoma.
- **Carbonic anhydrase 9 (CAIX) (CA9) protein expression** — Induced by VHL loss in clear-cell renal cell carcinoma and by hypoxia in carcinomas; no established expression role in uterine leiomyosarcoma.
- **Carcinoembryonic antigen-related cell adhesion molecule 5 (CEACAM5) protein expression** — Carcinoembryonic antigen of colorectal, gastric, pancreatic and lung adenocarcinoma; no reported expression in mesenchymal tumors.
- **Choline transporter-like protein 4 (SLC44A4) protein expression** — Up-regulated in pancreatic and colorectal adenocarcinoma; no reported expression in uterine smooth-muscle tumors.
- **Claudin-18 (isoform 2 / CLDN18.2) (CLDN18) protein expression** — The CLDN18.2 isoform is confined to gastric mucosa and its tumors, with further expression in pancreatic adenocarcinoma; no reported expression in uterine smooth-muscle tumors.
- **Delta-like protein 3 (DLL3) protein expression** — Neuroendocrine-lineage antigen of small cell lung cancer and high-grade neuroendocrine carcinoma; this histology has no neuroendocrine component.
- **Ectonucleotide pyrophosphatase/phosphodiesterase family member 3 (ENPP3) protein expression** —

Surface marker of clear-cell renal cell carcinoma and of basophils and mast cells; no reported expression in uterine smooth-muscle tumors.

- **Ephrin type-A receptor 2 (EPHA2) protein expression** — Over-expressed across esophageal, gastric, lung, breast, ovarian and glial tumors; no reported expression in uterine leiomyosarcoma, and the binder programs remain early.
- **Ephrin type-B receptor 4 (EPHB4) protein expression** — Reported on Kaposi sarcoma, mesothelioma and several carcinomas; the sarcoma evidence sits in vascular lineage rather than smooth muscle.
- **Epidermal growth factor receptor (EGFR) protein expression** — Driver or over-expressed target in lung, colorectal and head-and-neck carcinoma; it appears in this case record only as an exclusion criterion inside a lung-cancer trial and has no role in uterine leiomyosarcoma.
- **Epidermal growth factor receptor variant III (EGFRvIII) (EGFR (variant III)) protein expression** — Tumor-specific neoepitope confined to EGFR-amplified glioblastoma; no basis in a non-CNS mesenchymal tumor.
- **Epithelial cell adhesion molecule (EPCAM) protein expression** — Epithelial cell adhesion molecule, expressed across carcinomas and used to define epithelial lineage; a mesenchymal tumor is not expected to carry it.
- **Folate receptor alpha (FOLR1) protein expression** — Folate receptor alpha marks high-grade serous ovarian and endometrial carcinoma and lung adenocarcinoma; no reported expression in uterine leiomyosarcoma.
- **Frizzled-7 (FZD7) protein expression** — Marks Wnt-driven breast, gastric and Wilms tumor models; a preclinical target with no reported expression in uterine smooth-muscle tumors.
- **G-protein coupled receptor family C group 5 member D (GPRC5D) protein expression** — Plasma-cell-restricted antigen in multiple myeloma; no lineage basis in a soft-tissue sarcoma.
- **Glypican-3 (GPC3) protein expression** — Oncofetal antigen of hepatocellular carcinoma, with further expression in squamous lung and yolk-sac tumors; no reported expression in mesenchymal tumors of the uterus.
- **Guanylyl cyclase C (GUCY2C) protein expression** — Gut-restricted receptor expressed in colorectal and upper gastrointestinal adenocarcinoma; no reported expression in uterine smooth-muscle tumors.
- **Hepatitis A virus cellular receptor 1 (KIM-1 / TIM-1) (HAVCR1) protein expression** — Marks clear-cell renal carcinoma and a subset of ovarian clear-cell tumors; no reported expression in uterine leiomyosarcoma, and the panel lists no binder program behind it.
- **Hepatocyte growth factor receptor (c-MET) (MET) protein expression** — MET amplification, exon-14 skipping and protein over-expression define subsets of lung, gastric, liver and papillary renal cancer; no MET-addicted subset is described in uterine leiomyosarcoma.
- **Inactive tyrosine-protein kinase 7 (PTK7) protein expression** — Up-regulated in triple-negative breast, lung, ovarian and esophageal carcinoma; no reported expression in uterine leiomyosarcoma.
- **Inactive tyrosine-protein kinase transmembrane receptor ROR1 (ROR1) protein expression** — Onco-fetal receptor of chronic lymphocytic leukemia, mantle cell lymphoma and several carcinomas; no reported expression in uterine smooth-muscle tumors.
- **Integrin alpha-3 (ITGA3) protein expression** — Up-regulated on lung, gastric, pancreatic and bladder carcinoma with preclinical concepts only; no reported expression in uterine smooth-muscle tumors.
- **Integrin beta-6 (ITGB6) protein expression** — Alpha6beta6 is induced on epithelial tumors of the pancreas, head and neck, lung and cervix; it is an epithelial integrin with no reported role in mesenchymal tumors.
- **Interleukin-13 receptor subunit alpha-2 (IL13RA2) protein expression** — Strongly over-expressed in glioblastoma and little else; no basis in a non-CNS mesenchymal tumor.
- **Interleukin-3 receptor subunit alpha (CD123) (IL3RA) protein expression** — CD123 marks blastic plasmacytoid dendritic cell neoplasm and myeloid leukemia; no lineage basis in a soft-tissue sarcoma.
- **Kita-Kyushu lung cancer antigen 1 (KK-LC-1) (CT83) protein expression** — Cancer/testis antigen of

triple-negative breast, gastric, lung and cervical carcinoma, and its lead T-cell receptor is restricted to HLA-A*01:01, which this patient's confirmed typing does not include.

- **Leucine-rich repeat-containing G-protein coupled receptor 5 (LGR5) protein expression** — Wnt-high stem-cell marker in colorectal, gastric, ovarian and hepatocellular carcinoma with preclinical programs only; no reported expression in uterine leiomyosarcoma.
- **Leukocyte antigen CD37 (CD37) protein expression** — B-cell-lineage antigen targeted in lymphoma and acute myeloid leukemia; no lineage basis in a soft-tissue sarcoma.
- **Leukocyte immunoglobulin-like receptor subfamily B member 4 (LILRB4) protein expression** — Marker of monocytic acute myeloid leukemia and chronic myelomonocytic leukemia; no lineage basis in a soft-tissue sarcoma.
- **Ly6/PLAUR domain-containing protein 3 (C4.4A) (LYPD3) protein expression** — Over-expressed in lung and esophageal squamous carcinoma; no reported expression in uterine smooth-muscle tumors.
- **Lymphocyte antigen 6E (LY6E) protein expression** — Identified as a theranostic target in triple-negative breast cancer, with early-phase conjugate work only; no reported expression in uterine smooth-muscle tumors.
- **Matrix metalloproteinase-14 (MT1-MMP) (MMP14) protein expression** — Surface protease of invasive breast, pancreatic and head-and-neck carcinoma, with preclinical antibodies and conjugate concepts only; nothing here would act on a result in this histology.
- **Melanoma-associated antigen 1 (MAGEA1) protein expression** — Cancer/testis antigen of melanoma, lung, esophageal, ovarian and hematologic malignancy; it appears in this case record only as cross-tumor mechanistic support for how cancer/testis antigen expression is switched on, not as a target with expression evidence in this histology.
- **Membrane cofactor protein (CD46) (CD46) protein expression** — Over-expressed on myeloma plasma cells and prostate carcinoma; no reported expression in uterine leiomyosarcoma.
- **Mesothelin (MSLN) protein expression** — Mesothelin expression sits in mesothelioma and in pancreatic, ovarian and other epithelial tumors; despite its pan-cancer reputation it is not reported on uterine smooth-muscle tumors.
- **Metalloreductase STEAP1 (STEAP1) protein expression** — Expressed in prostate cancer and, among sarcomas, in Ewing sarcoma; that is a small round cell lineage driven by an EWSR1 fusion rather than smooth muscle.
- **Mucin-1 (MUC1-C / MUC1-CT cleavage product accessible on tumor surface) (MUC1) protein expression** — Aberrantly glycosylated mucin of breast, ovarian, pancreatic, lung and colorectal adenocarcinoma; an epithelial mucin with no reported expression in mesenchymal tumors.
- **Nectin-4 (NECTIN4) protein expression** — Lineage-restricted to urothelial carcinoma, with lesser expression in a few other carcinomas; no reported expression in uterine smooth-muscle tumors.
- **Neural cell adhesion molecule L1 (L1CAM) protein expression** — Re-expressed in neuroblastoma, glioma, ovarian and endometrial carcinoma; the uterine evidence is endometrial epithelium rather than smooth muscle. The L1 text matches elsewhere in this case record are line-of-therapy labels, not this target.
- **OX-2 membrane glycoprotein (CD200) (CD200) protein expression** — Up-regulated on chronic lymphocytic leukemia, myeloma and some melanoma and ovarian tumors; no reported expression in uterine smooth-muscle tumors.
- **Podocalyxin (PODXL) protein expression** — Sialomucin marking aggressive renal, ovarian, pancreatic and breast carcinoma; the panel lists no binder program, and there is no reported expression in uterine leiomyosarcoma.
- **Prostate stem cell antigen (PSCA) protein expression** — Up-regulated in prostate, pancreatic, gastric and bladder cancer; no reported expression in uterine smooth-muscle tumors.
- **Protein SSX2 (SSX2) protein expression** — Cancer/testis antigen whose sarcoma evidence is synovial sarcoma driven by the SS18-SSX fusion, a lineage and mechanism absent from uterine leiomyosarcoma.

- **Protein delta homolog 1 (DLK1) protein expression** — Oncofetal antigen of neuroblastoma, hepatoblastoma and rhabdomyosarcoma; the sarcoma evidence is skeletal-muscle and neural-crest lineage, with no reported expression in uterine smooth-muscle tumors.
- **Receptor tyrosine-protein kinase erbB-3 (ERBB3) protein expression** — HER3 expression evidence sits in lung, breast, head-and-neck and gastrointestinal carcinoma, and the plasma ERBB3 G914E variant on file sits at 0.9% variant allele fraction near the limit of detection, which is not a HER-family rationale.
- **SLIT and NTRK-like protein 6 (SLITRK6) protein expression** — Expressed on urothelial carcinoma and a fraction of lung and glial tumors; no reported expression in uterine leiomyosarcoma.
- **Seizure protein 6 homolog (SEZ6) protein expression** — Neuroendocrine-lineage antigen enriched in small cell lung cancer; this histology has no neuroendocrine component.
- **Sodium-dependent phosphate transport protein 2B (NaPi2b) (SLC34A2) protein expression** — NaPi2b is expressed in ovarian and lung adenocarcinoma and thyroid cancer with apically restricted normal expression; no reported expression in mesenchymal tumors.
- **Somatostatin receptor type 2 (SSTR2) protein expression** — Somatostatin receptor 2 defines well-differentiated neuroendocrine tumors and underpins their radioligand therapy; this is a high-grade smooth-muscle sarcoma with no neuroendocrine differentiation.
- **Tetraspanin-8 (TSPAN8) protein expression** — Up-regulated in colorectal, pancreatic, gastric and hepatocellular carcinoma with preclinical antibodies only; no reported expression in uterine smooth-muscle tumors.
- **Tissue factor (F3) protein expression** — Tissue factor is the target of an approved conjugate in cervical carcinoma and is reported across several other carcinomas; no expression basis in uterine smooth-muscle tumors.
- **Tumor necrosis factor receptor superfamily member 10B (DR5) (TNFRSF10B) protein expression** — Death receptor 5 agonists are in development against carcinomas and are not selected by an expression assay in the way the rest of this panel is; no expression result here would change what is available.
- **Tumor-associated calcium signal transducer 2 (Trop-2) (TACSTD2) protein expression** — Trop-2 is broadly over-expressed across epithelial tumors and carries approved conjugates in breast, lung and urothelial cancer; broad as that is, it stays within epithelial lineage and is not reported on uterine smooth-muscle tumors.
- **Tyrosine-protein kinase Mer (MERTK) protein expression** — Aberrantly expressed on leukemia, glioblastoma, melanoma and lung cancer cells and on tumor-promoting macrophages, with preclinical programs only; no reported expression in uterine leiomyosarcoma.